

(a) Sparse Matrix M

	0	1	2	3	4	5	6	7
0	.	5	.	.	.	2	.	.
1	.	.	.	4	.	.	.	.
2	7	.	.	.	.	.	.	.
3	.	.	.	.	.	.	8	3

6 non-zeros in 32 cells

(c) SpMV Kernel

```
1 ...
2 for (i = 0; i < M; i++) {
3     y[i] = 0;
4     for (j = row_off[i];
5         j < row_off[i+1]; j++) {
6         y[i] += A_val[j] *
7             x[A_col[j]];
8     }
9 }
```

(b) CSR Encoding

row_off[]	0	2	3	4	6	
row start indices						
A_col[]	1	5	3	0	6	7
column indices						
A_val[]	5	2	4	7	8	3
non-zero indices						

(d) Runtime Access Pattern

